THE DERIVED CATEGORY OF QUASI-COHERENT MODULES ON A QUASI-COMPACT QUASI-SEPARATED SCHEME IS A DERIVED CATEGORY OF A DIFFERENTIAL GRADED ALGEBRA

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Allowed standard prerequisites (editorial).

- A quasi-compact quasi-separated scheme admits a perfect K-injective complex that generates the derived category of quasi-coherent modules, by the Bondal-van den Bergh theorem and the existence of K-injective resolutions. invoked at proof lines 4690-4693. Provides the generator K^bullet out of which the dg algebra is built.
- For a K-injective complex the cohomology of the dg endomorphism algebra computes the Ext groups of the corresponding object. invoked at proof lines 4701-4706. Identifies H^n(E) with Ext^n(K^bullet, K^bullet), whose finiteness follows from the vanishing lemma for Ext out of perfect objects into bounded quasi-coherent complexes.
- The dg tensor functor - tensor^L_E K^bullet and its right adjoint R Hom(K^bullet, -), with the fully faithfulness criterion in the compact case and the equivalence criterion for a fully faithful functor whose adjoint has vanishing kernel. invoked at proof lines 4715-4727 and proof lines 4737-4739. Assembles the equivalence from the adjunction.

Source reference. Original theorem 09IS: sources/stacks-perfect/pinned-original-source.tex, source lines 4111–4111. Cited at proof lines 4692-4693 for the existence of K^bullet; the construction of (E, d) from it is inside the counted span.

Source reference. Original section 09LU: sources/stacks-chow/tags.txt, source lines 12157–12157. Cited at proof line 4700 as the reference section for the dg category conventions; the definition of (E, d) is inside the counted span.

Source reference. Original lemma 09M4: sources/stacks-perfect/pinned-original-source.tex, source lines 3465–3465. Cited at proof lines 4707-4708 for finiteness of the nonzero cohomology groups of E; the Ext identification is inside the counted span.

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Source reference. Original lemma 09LX: sources/stacks-chow/tags.txt, source lines 12160–12160. Cited at proof lines 4715-4716 to define the tensor functor; the fully faithfulness argument is inside the counted span.

Source reference. Original proposition 09M1: sources/stacks-perfect/pinned-original-source.tex, source lines 4464–4464. Cited at proof lines 4717-4718 for the compactness of K^bullet; the fully faithfulness conclusion is inside the counted span.

Source reference. Original lemma 09LZ: sources/stacks-chow/tags.txt, source lines 12162–12162. Cited at proof lines 4721-4722 for full faithfulness; the compactness input and the Ext identification are inside the counted span.

Source reference. Original lemma 09LY: sources/stacks-chow/tags.txt, source lines 12161–12161. Cited at proof lines 4726-4727 for the right adjoint; the quasi-inverse and kernel arguments are inside the counted span.

Source reference. Original lemma 09M3: sources/stacks-perfect/pinned-original-source.tex, source lines 4650–4650. Cited at proof line 4730 to land the tensor functor in D_QCoh; the kernel vanishing by generation is inside the counted span.

Source reference. Original lemma 09J1: sources/stacks-chow/tags.txt, source lines 12059–12059. Cited at proof lines 4738-4739 to conclude the equivalence; the kernel vanishing by generation is inside the counted span.

Theorem 0.2. *Let X be a quasi-compact and quasi-separated scheme. Then there exist a differential graded algebra (E, d) with only a finite number of nonzero cohomology groups $H^i(E)$ such that $D_{QCoh}(\mathcal{O}_X)$ is equivalent to $D(E, d)$.*

Proof. Let $K^\bullet$ be a K-injective complex of $\mathcal{O}$-modules which is perfect and generates $D_{QCoh}(\mathcal{O}_X)$. Such a thing exists by Theorem 09IS and the existence of K-injective resolutions. We will show the theorem holds with

$$(E, \mathrm{d}) = \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{O}_X)}(K^\bullet, K^\bullet)$$

where $\mathrm{Comp}^{dg}(\mathcal{O}_X)$ is the differential graded category of complexes of $\mathcal{O}$-modules. Please see Differential Graded Algebra, Section 09LU. Since $K^\bullet$ is K-injective we have

$$H^n(E) = \mathrm{Ext}^n_{D(\mathcal{O}_X)}(K^\bullet, K^\bullet) \tag{0.2.1}$$

for all $n \in \mathbf{Z}$. Only a finite number of these Exts are nonzero by Lemma 09M4. Consider the functor

$$- \otimes^{\mathbf{L}}_E K^\bullet : D(E, \mathrm{d}) \longrightarrow D(\mathcal{O}_X)$$

of Differential Graded Algebra, Lemma 09LX. Since $K^\bullet$ is perfect, it defines a compact object of $D(\mathcal{O}_X)$, see Proposition 09M1. Combined with (0.2.1) the functor above is fully faithful as follows from Differential Graded Algebra, Lemmas 09LZ. It has a right adjoint

$$R\,\mathrm{Hom}(K^\bullet, -) : D(\mathcal{O}_X) \longrightarrow D(E, \mathrm{d})$$

by Differential Graded Algebra, Lemmas 09LY which is a left quasi-inverse functor by generalities on adjoint functors. On the other hand, it follows from Lemma 09M3 that we obtain

$$- \otimes^{\mathbf{L}}_E K^\bullet : D(E, \mathrm{d}) \longrightarrow D_{QCoh}(\mathcal{O}_X)$$

and by our choice of $K^\bullet$ as a generator of $D_{QCoh}(\mathcal{O}_X)$ the kernel of the adjoint restricted to $D_{QCoh}(\mathcal{O}_X)$ is zero. A formal argument shows that we obtain the desired equivalence, see Derived Categories, Lemma 09J1. $\square$